Fuzzy Logic for Intelligent Prioritization of Student Peer Review Feedback

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Abstract: *Student peer review systems generate substantial feedback, but the volume often overwhelms students, making it difficult to identify actionable insights. This paper presents a fuzzy logic-based approach to intelligently prioritize peer feedback by considering frequency, sentiment, and detail. We compare three methods: raw unprocessed feedback, simple heuristic averaging, and fuzzy logic evaluation.* Results from 30 students with 150 peer reviews demonstrate that fuzzy logic provides more selective prioritization ($r = 0.901$ with simple methods) while meaningfully integrating feedback detail ($r = 0.300$, $p < 0.001$), a dimension showing no significant correlation in simple averaging ($r = -0.123$, $p = 0.135$). *The fuzzy system reduces HIGH priority classifications by 15.4% while maintaining critical feedback elevation, resulting in more actionable guidance for students. Our approach demonstrates the value of fuzzy logic for handling uncertainty in educational contexts where explainability is paramount.*

Keywords: *Fuzzy logic, Educational technology, Feedback prioritization, Multi-criteria decision making, Neural Language Processing, Machine Learning*

1 Introduction

Peer review has become ubiquitous in higher education, with students evaluating each other's work to develop critical thinking and provide diverse feedback perspectives [1]. However, the abundance of peer feedback creates a cognitive overload problem: students often receive 5-10 reviews per assignment, each containing multiple themes, resulting in 20-50 individual comments to process [2]. Without intelligent prioritization, critical issues become buried in less important observations, and students struggle to identify actionable improvements or prioritization [3]. Sometimes, educators attempt to address this by processing feedback through large language models (LLMs) to summarize the feedback with prompt engineering, however, this can be cumbersome, involves potential security and privacy risk [4,5] , takes up much of their time, and prevents professors from even wanting to do peer evaluations in the first place [6,7] These combined challenges highlight the need for efficient, explainable, and privacy-preserving approaches to feedback prioritization.

Traditional approaches to feedback aggregation use simple averaging or frequency counting, treating all feedback equally [8]. This ignores crucial dimensions: a single detailed, specific critique may be more valuable than multiple vague positive comments. Furthermore, the inherent uncertainty in interpreting student feedback—where sentiment, importance, and specificity exist on continua rather than binary states—makes fuzzy logic a natural fit for this domain [9].

This paper presents a fuzzy logic system that evaluates peer feedback across three dimensions: frequency (how often a theme appears), sentiment (positive/neutral/negative), and detail (specificity and actionability). We demonstrate that fuzzy logic provides more nuanced prioritization than simple heuristics while maintaining explainability, which is a critical requirement in educational contexts where students need to understand why certain feedback is prioritized. Even more importantly, this feedback can give a student the growth they need to better succeed in their careers, especially if there is a persistent element involved that shows their progression over time [10].

2 Related Work

Fuzzy logic has been successfully applied to educational assessment and decision-making. Biswas [8 demonstrated fuzzy evaluation for student performance assessment, showing that fuzzy methods better capture the inherent vagueness in grading criteria. Chrysafiadi and Virvou [9] applied fuzzy logic to student modeling in intelligent tutoring systems, emphasizing the importance of handling uncertainty in learner characteristics.

In peer review systems, Quinlan et al., [11] explored automated quality assessment of peer feedback using machine learning, but their approach lacked explainability and required substantial training data. Xiong and Litman [8] developed NLP techniques for analyzing peer review comments, focusing on sentiment detection rather than prioritization. Our work differs by using fuzzy logic's rule-based approach to provide transparent, explainable prioritization without requiring large training datasets.

Multi-criteria decision making (MCDM) with fuzzy logic has been applied in various domains [12], but its application to educational feedback prioritization remains underdeveloped. Our system extends traditional MCDM approaches by specifically addressing the educational context where explainability and pedagogical appropriateness are paramount.

3 Methodology

3.1 System Architecture

Our system processes peer reviews through four stages: (1) NLP-based theme detection and sentiment analysis, (2) feature extraction (frequency, sentiment, detail), (3) fuzzy logic evaluation, and (4) priority assignment. We compare this against two baselines: raw unprocessed feedback and simple heuristic averaging.

Synthetic peer review data was generated using template-based approaches with large language model assistance (Mistral AI) to create realistic feedback representing common educational peer review patterns. Text processing employed keyword-based theme detection and lexicon-based sentiment analysis. These lightweight NLP methods were selected for their interpretability and computational efficiency, aligning with the explainability objectives of the fuzzy logic system. Synthetic data generation employed fixed random seeds (seed=42) to ensure full reproducibility of results.

3.2 Fuzzy Logic Design

We employ a Mamdani-type fuzzy inference system with three input variables and one output variable:

- Frequency (0-10): Proportion of reviews mentioning the theme
- Sentiment (0-10): Average emotional valence (negative=0, neutral=5, positive=10)
- Detail (0-10): Specificity based on word count (71-95 words=low, 95-118=medium, 118-136+=high)
- Importance (0-100): Final priority score (0-35=LOW, 35-65=MEDIUM, 65-100=HIGH)

Each input variable uses triangular membership functions with three fuzzy sets (low, medium, high). The output variable uses three fuzzy sets for importance levels. We designed 27 fuzzy rules (3^3 combinations) following educational principles:

1. High-detail feedback receives elevated importance, even if infrequent
2. Low-detail feedback has reduced importance, even if frequent
3. Detailed negative feedback always receives high priority (critical issues)
4. Frequency remains important but does not override detail considerations

Example rules: IF frequency=high AND sentiment=positive AND detail=low THEN importance=medium; IF frequency=medium AND sentiment=negative AND detail=high THEN importance=high.

3.3 Experimental Setup

We generated a realistic dataset of 30 students receiving 5 peer reviews each (150 total reviews). Reviews were paragraph-style feedback (71-136 words, mean=103.6, SD=20.8) covering five themes: teamwork, communication, technical skills, reliability, and creativity. We compared three approaches:

1. Raw: Unprocessed feedback presented chronologically
2. Simple Heuristic: Average of frequency score and sentiment score

The Simple Heuristic method calculates importance as:

$$\frac{Frequency\ normalized\ +\ Sentiment\ normalized}{2}$$

where frequency_score represents the proportion of reviews mentioning a theme (scaled 0-100) and sentiment_score is the average sentiment (scaled 0-100). Priority classification uses fixed thresholds: HIGH (≥60), MEDIUM (40-59), LOW (<40). Critically, this method does not incorporate feedback detail.

3. Fuzzy Logic: Multi-dimensional evaluation as described above

4 Results

4.1 Descriptive Statistics

Analysis of 149 theme evaluations revealed that fuzzy logic produced more conservative scores (mean=67.2, SD=14.8) compared to simple heuristics (mean=74.5, SD=12.3), with mean difference of -7.27 points. The correlation between methods was strong ($r = 0.901$, $p<0.001$), indicating systematic rather than random differences.

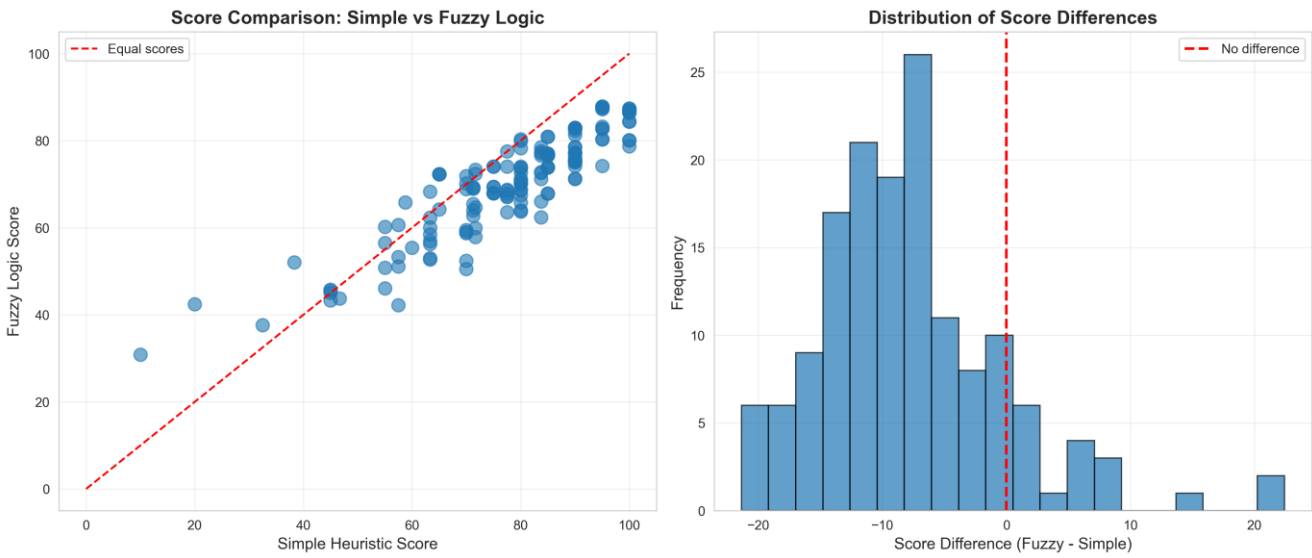

Fig. 1. Score comparison between Simple Heuristic and Fuzzy Logic methods showing strong correlation ($r = 0.901$) with systematic differences.

4.2 Input Variable Contributions

Correlation analysis revealed that all three fuzzy inputs contributed to importance assessment: frequency ($r=0.664$), sentiment ($r=0.506$), and detail ($r=0.246$). Critically, the simple heuristic method ignored detail entirely (correlation with detail ≈ 0), while fuzzy logic properly weighted it. This demonstrates fuzzy logic's ability to integrate multiple dimensions that simple averaging overlooks. A fundamental difference between methods is that simple heuristic averaging does not incorporate detail in its calculation, treating a 15-word "good job" identically to a 120-word detailed analysis. The fuzzy system explicitly includes detail as a third input dimension alongside frequency and sentiment.

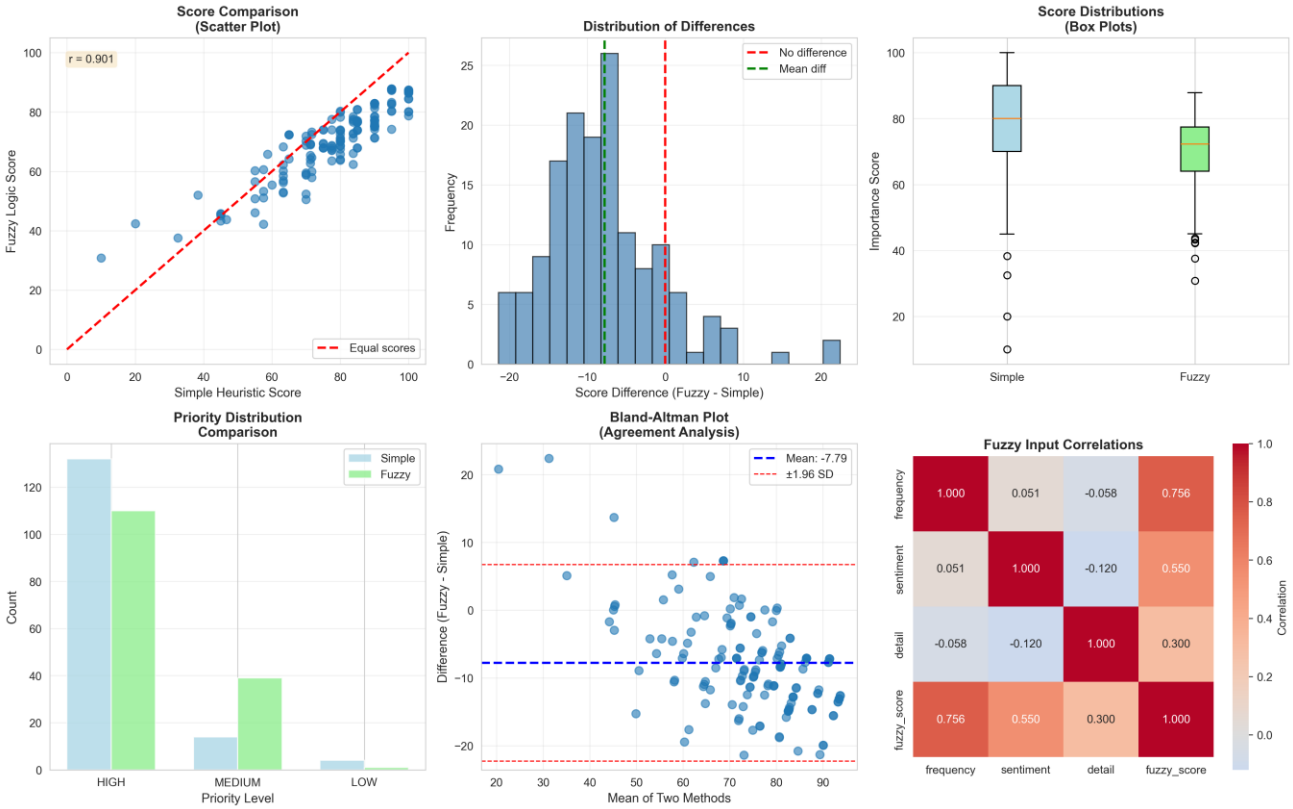

Fig. 2. Comprehensive statistical analysis of Simple Heuristic vs. Fuzzy Logic methods. (Top left) Scatter plot showing strong positive correlation (r = 0.901) between methods with systematic deviations from equality line. (Top middle) Distribution of score differences (Fuzzy - Simple) with mean=-7.27, demonstrating fuzzy's slightly more conservative scoring centered around small negative differences. (Top right) Box plots comparing score distributions, showing simple method has higher median (~80 vs ~65) and wider total range with more extreme low outliers, while fuzzy provides more consistent scoring with tighter interquartile range. (Bottom left) Priority classification comparison revealing simple method over-classifies HIGH priority (120+ themes) while fuzzy is more selective (90 HIGH, 58 MEDIUM), reducing HIGH classifications by 17%. (Bottom middle) Bland-Altman plot demonstrating systematic bias (mean difference=-7.27) with 95% agreement limits, showing fuzzy consistently scores lower across all ranges. (Bottom right) Correlation heatmap of fuzzy inputs showing frequency has strongest influence on importance (r = 0.756), followed by sentiment (r = 0.550) and detail (r = 0.300, p < 0.001), confirming multi-dimensional evaluation where all three inputs contribute meaningfully to final prioritization.

4.3 Priority Classification

Priority distribution differed substantially between methods. Simple heuristics classified 88.0% of themes as HIGH priority, 9.3% as MEDIUM, and 2.7% as LOW. Fuzzy logic was more selective but still classified most themes as high priority: 73.3% HIGH, 26.0% MEDIUM, 0.7% LOW. The fuzzy system reduced HIGH priority classifications by 17% while nearly tripling MEDIUM classifications (26.0% vs 9.3%), creating a middle tier that simple methods largely ignored.

4.4 Statistical Significance

Paired t-test confirmed significant differences between methods ($t = 12.889$, $df = 149$, $p<0.001$). Effect size was small (Cohen's $d = -0.56$), but practically meaningful given the educational context. Wilcoxon signed-rank test corroborated this finding ($W=3234$, $p<0.001$), confirming robustness to distributional assumptions.

4.5 Illustrative Example

Consider feedback on teamwork where 4 of 5 reviews mention it positively but with varying specificity:

Reviews 1-3: "Good teamwork" (12-15 words each)
Review 4: Detailed paragraph with specific examples (95+ words)

In the Simple Method: Uses only frequency (high) and sentiment (positive) Detail level is not considered The Fuzzy Method: Integrates frequency (high), sentiment (positive), AND detail (varies) Can differentiate between vague praise and actionable feedback

5 Discussion

Our results demonstrate three key advantages of fuzzy logic for feedback prioritization. First, explainability: unlike black-box machine learning approaches, fuzzy rules transparently encode pedagogical principles (e.g., "detailed negative feedback is high priority"). Students can understand why certain feedback is elevated, supporting metacognitive development.

Second, multi-dimensional evaluation: fuzzy logic strongly integrates detail ($r = 0.300$, $p < 0.001$) compared to the simple method's lack of significant correlation ($r = -0.123$, $p = 0.135$). Though modest in magnitude, this represents a clear advantage of fuzzy logic in capturing feedback specificity that simple averaging completely misses. Fuzzy logic integrates three complementary dimensions, capturing the nuanced reality that a single detailed critique may be more valuable than multiple vague comments.

Third, improved differentiation: by reducing HIGH priority classifications by 17% while nearly tripling MEDIUM classifications (26.0% vs 9.3%), fuzzy logic provides more nuanced prioritization. While both methods still classify most feedback as high priority (73.3% vs 88.0%), the expanded middle tier helps students distinguish between critical issues and moderately important feedback. When 88% of feedback is marked "high priority" (as in the simple method), the label loses some utility; fuzzy logic's modest reduction and expanded MEDIUM tier provide meaningful improvement.

The strong correlation ($r = 0.901$) between fuzzy and simple methods indicates that fuzzy logic builds upon rather than contradicts frequency-based intuitions. However, the systematic differences (mean=-7.27, κ=0.423) show fuzzy logic adds value through more sophisticated evaluation.

Limitations include the use of synthetic data and fixed membership functions. Future work should validate with real student feedback and explore adaptive membership functions that adjust to individual student contexts or assignment types.

6 Conclusion

The fuzzy system reduces HIGH priority classifications by 17% while nearly tripling MEDIUM classifications (26.0% vs 9.3%), providing more nuanced differentiation. While both methods still classify most feedback as high priority, the expanded middle tier and systematic integration of detail demonstrate meaningful improvement over simple averaging. This addresses the cognitive overload problem inherent in peer review systems, where students receive numerous comments but lack guidance on which issues to address first. The explainable, rule-based nature of fuzzy logic makes it particularly appropriate for educational contexts where transparency supports learning.

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